



MID-TERM PROJECT

Beyond RFM?

Evaluating Product-Aware Household Segmentation in Grocery Retail

Grocery Store Dataset (Transaction & Product)

Course: Data Structures and Algorithms with Python
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RESEARCH AT A GLANCE

Does more shopping detail produce better customer segments?

We tested an intuitive expectation: adding basket behavior and department preferences should reveal more useful household groups. The richer representation did not clearly improve the original stability and later-spending criteria.

2.60 million transaction lines $\longrightarrow$ **2,498 households**

Days 1–500: build and freeze groups | Days 501–711: describe later behavior

01 RFM

Recency, frequency and monetary value
3 features

02 Behavior

RFM + basket size, variety and discount indicators
8 features

03 Product

Behavior + department spending shares
17 features

Three results that change the recommendation

Question	Evidence	Interpretation
Will groups repeat?	Subsample ARI: **0.980 RFM**, 0.964 Product	RFM has higher observed membership stability.
Do groups separate later spend?	Explained variance: **20.4% RFM**, 20.2% Product	No clear gain from enrichment at $k = 2$.
Does weighting matter?	At $k = 3$, ARI to default falls to **0.433 / 0.460** at half / double category weight.	Segments depend materially on how similarity is defined.

Decision. Use RFM as the default for broad activity monitoring under the original criteria. Evaluate department preferences against a category-specific objective before assigning them operational value.

See the mechanism. The household case studies show who changes groups; the membership-flow diagram makes weighting sensitivity visible. The exploratory later-category check finds some persistent differences, but only a small, uncertain Product advantage (0.386% versus 0.299% of share variation explained).

These are observational comparisons, not evidence of campaign lift. The later-category extension was added after inspecting the original holdout results; it is exploratory.

Abstract

This study evaluates whether household shopping behavior and department composition improve grocery retail segmentation beyond a recency, frequency and monetary-value (RFM) baseline. Audited transaction and product records yield 2,498 calibration households. K-means, Ward hierarchical clustering and DBSCAN are compared using within-space geometry, cluster size, sampling stability, fixed-count ablation, common-space diagnostics and a later-period descriptive holdout. All representative K-means fits select two broad activity groups. K-means is more stable than Ward; RFM's mean subsample adjusted Rand index is 0.980, versus 0.964 for the product-aware representation. Later-spend explained variance is 0.204 for RFM and 0.202 for Product, and the fixed three-cluster comparison likewise does not favor enrichment. Product features supply department descriptions, but membership is sensitive to their weight. After label alignment, 287 households change groups between RFM and Product. An exploratory later-category extension finds a small, uncertain Product advantage in department-share separation (0.386% versus 0.299% of variation explained). RFM remains the preferred default under the tested design. The results describe observational purchase patterns and do not establish causal marketing lift.

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1 Problem and decision context

Would two households with similar spending belong in different segments because they buy different products? This is the practical choice behind adding category preferences to an RFM baseline.

Grocery retailers observe more than the amount and timing of household spending: they also observe which departments households buy from and how they assemble baskets. A practical segmentation should compress this behavior into a small number of understandable groups. The additional detail is useful only if it produces a defensible, stable partition and an interpretation that supports a clear decision. Complexity alone is not evidence of improvement.

This study asks whether product-aware information improves household segmentation relative to a recency, frequency and monetary-value (RFM) baseline in dunnhumby's *The Complete Journey*. The assignment's transaction and product clustering option is retained. There is no supervised target variable or known correct household label. The analysis therefore evaluates the geometry, robustness and descriptive meaning of alternative partitions, rather than reporting classification accuracy.

The research questions are: (1) How do baseline and enriched representations partition the same household cohort? (2) Are the partitions reproducible under different initializations and household samples? (3) Do their profiles remain understandable when assessed with common measurements? (4) Do calibration-defined groups show distinguishable behavior in a later period? These questions are related but cannot be collapsed into a single score. Clustering depends on the representation, similarity measure and intended use [7].

A credible negative result is possible. If richer representations are less stable, create impractically small groups, or fail to add useful distinctions, the simpler baseline remains preferable. Any operational recommendation in this report is provisional: the dataset supplies observational purchase histories, not randomized evidence that a segmentation increases sales or promotion response.

2 Data provenance and analytical unit

The assignment points to the Kaggle distribution of dunnhumby's *The Complete Journey*. The original provider describes a longitudinal record of frequently shopping households, with transactions and supporting product, demographic and marketing tables [1]. The local audit determines the actual available files and observation range rather than assuming every public distribution is identical.

The analytical unit is a household with at least one eligible calibration transaction. Transaction lines are aggregated to baskets and households; product metadata links purchased product identifiers to departments. Household identifiers are grouping keys, never numerical predictors. Basket identifiers identify shopping occasions. Day indices express relative observation time; they should not be presented as independently verified calendar dates.

The cohort is selective. Observed households are established shoppers in a particular historical retail setting, and their activity at other retailers is not captured. Absence of a later recorded transaction is therefore described as *no observed later purchase*, not proof that a household churned. Demographic availability is incomplete and need not be representative. The study avoids making demographic or individual-sensitive claims.

3 Design principles and safeguards

3.1 Temporal separation

Calibration transactions alone determine the household cohort, feature values, fitted preprocessing and cluster assignments. Later transactions are reserved for descriptive evaluation. This prevents later spend-

ing from influencing the partition it is used to describe. The temporal design is a retrospective holdout, not a prospective deployment experiment. Standard preprocessing safeguards are applied to keep fitting and evaluation roles distinct [2].

3.2 Representations and fair comparison

The RFM representation summarizes time since the most recent observed purchase, number of shopping occasions, and observed spending during calibration. Behavioral additions capture shopping intensity and basket characteristics. Product-aware additions summarize department preferences. Definitions, dimensions and transformations are reported with the generated feature tables below.

Features on different scales can dominate Euclidean distances without standardization. Skewed non-negative activity variables receive a documented transformation before scaling. Department shares are compositional: their sum is constrained, and redundant categories and differing numbers of columns can alter distance weighting. Product preferences must therefore be interpreted as a different representation of similarity, not an automatic refinement of a universal ground truth.

All feature sets use the same household rows, temporal boundary and candidate cluster-count range. A paired comparison at fixed cluster count distinguishes changes in representation from changes in model size. Internal metrics are also supplemented with assessments in a common RFM space, profile summaries in original units, and agreement between partitions. These are complementary views; a common RFM space itself favors the baseline's chosen notion of similarity.

3.3 Algorithms and evaluation

K-means minimizes within-cluster squared Euclidean distance to cluster centroids [4]. Repeated initialization reduces sensitivity to an unlucky starting configuration, but does not establish a unique natural partition. The elbow curve shows diminishing improvements in this objective as more clusters are added. Inertia is comparable across cluster counts within a fixed representation; its numerical scale changes with the representation.

For household i, silhouette is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}},$$

where $a(i)$ is mean distance to the household's assigned group and $b(i)$ is the smallest mean distance to another group [5]. Values close to one indicate compact separated assignments, near zero indicate boundaries, and negative values suggest another group is closer on average. The implementation uses a deterministic sample of up to 1,500 eligible households (random seed 42) to estimate average silhouette. Silhouette depends on the underlying distance space. Comparing its values across feature sets cannot, on its own, rank the business quality of those feature sets.

DBSCAN supplies a density-based contrast to centroid clustering [3]. It can label observations as noise and may fail to yield a useful multi-group partition at a given neighborhood radius. Scores calculated only on non-noise rows must be read together with coverage: excluding difficult observations can make a score look favorable. A failed or low-coverage configuration is a result to report, not a reason to silently remove that candidate.

Adjusted Rand index (ARI) quantifies pairwise agreement between two partitions, adjusted for chance; one means identical assignments up to label permutation. Seed stability tests optimization sensitivity. Subsample stability additionally perturbs which households inform the centroids. Stable clustering need not be substantively correct, and instability may reflect weak structure, model choice or sampling effects [6]. The number of repetitions, sampling fraction and variability must accompany averages.

3.4 Selection and interpretation

The final comparative interpretation weighs cluster sizes, within-space geometry, fixed-count ablation, stability and intelligible profiles. A score-maximizing partition with a tiny outlier group may be less useful than a balanced, stable one. Cluster numbers are arbitrary identifiers, and descriptive labels summarize profiles rather than define fixed household types. Later-period summaries assess persistence and differentiation, with both means and distributions considered where available. They do not provide classification accuracy, predict individual outcomes with calibrated error, or identify the incremental effect of a marketing action.

4 Feature construction

Table 1: Household representations fitted on calibration purchases.

Representation	Definitions and preprocessing
RFM (3 columns)	Recency is $501 - \max(\text{DAY})$; frequency is distinct basket count; monetary value is positive recorded sales summed over calibration. Apply $\log(1 + x)$, then standardize each column.
Behavior (8 columns)	RFM plus mean basket sales, mean basket line count, distinct purchased products, distinct departments and proportion of lines carrying any negative retail/coupon discount field. Log-transform all except the discount proportion, then standardize.
Product (17 columns)	Behavior plus sales shares in the eight largest calibration departments and an Other category. Take square roots, center each share coordinate, and scale the composition block to the same total variance as the eight-column Behavior block.

The eight largest departments are selected from calibration sales, not the later period. The Other group retains the remaining sales so the composition sums to one. Square-root shares provide Hellinger-style coordinates: absent departments remain finite without adding arbitrary pseudocounts. Centering does not change pairwise distances; the block scale explicitly controls how much department composition contributes relative to behavior. Equal block variance is a modeling choice, not an estimated economic optimum. Sensitivity to this weight is an important extension.

Frequency measures distinct baskets rather than transaction-line count. Mean basket lines counts recorded positive purchase lines rather than distinct items or units. Product breadth and department breadth measure diversity over the whole calibration period, so they remain related to shopping intensity. Mean basket value is also algebraically connected to total sales and basket frequency. Standardization does not remove those relationships. The enriched representation may therefore still be dominated by purchase intensity; the results must demonstrate whether it adds useful preference distinctions.

Retail and coupon discount columns contain negative values in this dataset. They are not treated as invalid purchases or subtracted from recorded sales; the study uses their sign only as a recorded-discount indicator. Positive-purchase filtering defines the estimand: returns and zero-price transactions are outside the clustering analysis. This is appropriate for describing recorded buying activity but does not estimate net household profitability.

5 Reproducible analytical workflow

The executable notebook embeds the analysis implementation, mirrors it in the project source module, and regenerates all reported tables and figures. The workflow first verifies the two required input files, loads transactions and products, checks product-key uniqueness, and performs a validated many-to-one left join. The joined row count must match the transaction count. It records unmatched products, missing values, nonpositive quantities and sales, duplicate source rows, and basket consistency checks.

Exact duplicate source rows are retained and counted: repeated item scans cannot be distinguished reliably from erroneous duplication without transaction-line identifiers. This choice differs from accidental duplicates introduced by a join; the latter are prevented by key validation and row-count checks. Missing departments, including blank text, receive an explicit Unknown category. Other exclusion and audit counts are reported in the following section.

The same cohort and derived matrices feed K-means, Ward hierarchical clustering and DBSCAN. The K-means and Ward grid uses two through eight clusters. K-means uses 20 initializations for the main fit with random seed 42. DBSCAN uses a minimum of ten samples and neighborhood radii chosen from the 50th, 70th, 85th and 95th percentiles of tenth-neighbor distances (including the observation itself among the ten points) in each space. These are exploratory settings, not a comprehensive density-model optimization.

A representative K-means fit is chosen separately within each space by highest within-space silhouette among partitions with every group containing at least five percent of the cohort. This rule selects a representative model within a space; it does not declare the highest silhouette across spaces a winner. Repeated-initialization tests use 20 one-initialization fits. Subsample tests use 20 paired independent samples, each containing 80 percent of households, and compare assignments only for shared households. Preprocessing is held fixed, so these checks assess conditional partition sensitivity and understate full-pipeline uncertainty.

Later positive purchase sales and distinct baskets are summarized for every calibration household. Households without later recorded purchases receive zero, retaining the cohort rather than conditioning on return shoppers. Mean-spend uncertainty uses 1,000 within-cluster bootstrap resamples, giving percentile intervals conditional on the fitted groups. These are not simultaneous confidence intervals and do not incorporate cluster-selection uncertainty. Between-group explained variance (η^2) is descriptive, can change with group count, and must be read alongside the paired fixed-count comparison.

6 Audit and exploratory findings

Table 2: Observed source and analytical sample.

Quantity	Count / range
Raw transaction lines	2,595,732
Product records	92,353
Observed households	2,500
Observed day indices	1–711
Excluded nonpositive quantity or sales lines	18,917
Retained positive purchase lines	2,576,815
Calibration lines (day ≤ 500)	1,726,165
Later lines (day > 500)	850,650
Calibration household cohort	2,498
Later-only households outside cohort	2
Cohort households with no later purchase	59

The transaction table has 12 columns and the product table seven. Both required tables contain zero missing cells under the CSV parser, no exact duplicate transaction rows and no duplicate product keys. All purchased product identifiers match the product table; the join preserves the transaction row count. Basket identifiers have neither conflicting household assignments nor conflicting day assignments. The eight-file schema inventory also verifies UTF-8-compatible decoding of the complete byte streams and records column names and sampled types. Only transactions and products enter clustering; demographics and campaign tables are intentionally not joined. These checks support the household aggregation and

rule out join multiplication in this run. They do not prove that source measurement is error-free.

There are 14,466 nonpositive-quantity lines and 18,850 nonpositive-sales lines; their union is 18,917, or 0.73% of raw lines. These counts overlap and must not be added. Negative retail discounts occur on 1,303,026 lines and are retained and used only to construct the negative-discount indicator. The calibration covers days 1–500; the later period covers days 501–711. Different window lengths mean calibration and later totals should not be compared as direct growth rates.

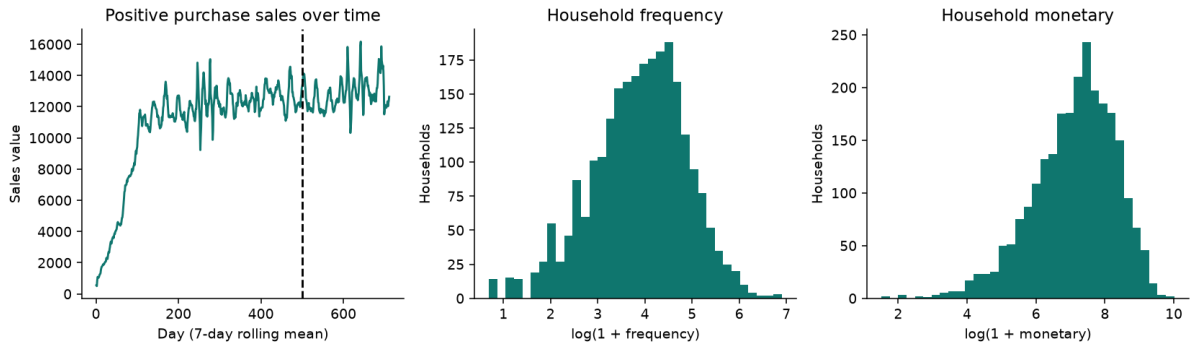


Figure 1: Recorded positive-purchase sales and household activity distributions. The dashed temporal boundary marks the calibration/later split on the exact day scale. Household activity is displayed after a log transform because raw purchase intensity is strongly heterogeneous.

The EDA motivates a baseline focused on shopping intensity and a transformed distance space. Monetary value and frequency are related because households with more shopping occasions have more opportunities to spend. Basket-level and assortment measures can clarify that relationship, but may also repeat the same underlying intensity signal. Department shares provide the most direct attempt to add preference information. The selected eight departments are Grocery, Drug GM, Produce, Meat, Kiosk-Gas, Meat-Packaged, Deli and Pastry.

7 Model comparisons

7.1 Within-space selection

All three representative K-means models select two clusters under the predeclared minimum-size and silhouette rule. The RFM groups contain 882 and 1,616 households; Behavior contains 809 and 1,689; Product contains 843 and 1,655. These are substantial groups rather than isolated outlier collections.

Table 3: Selected K-means models. Silhouette and Davies–Bouldin (DB) values are measured in each model's own space; they do not establish a cross-space ranking.

Space	Dimensions	k	Smallest group	Silhouette	DB
RFM	3	2	882	0.456	0.857
Behavior	8	2	809	0.329	1.237
Product	17	2	843	0.206	1.984

Two groups offer the strongest within-space K-means separation among eligible candidates. The shapes of the curves and the elbow diagnostic are supporting evidence, not proof that grocery households intrinsically fall into two discrete types. A small number of clusters can summarize a continuous intensity gradient.

The tested DBSCAN grid does not produce a compelling full-cohort alternative. In RFM at the 70th-percentile neighborhood setting, two groups remain after 18.2% of households are marked noise, with

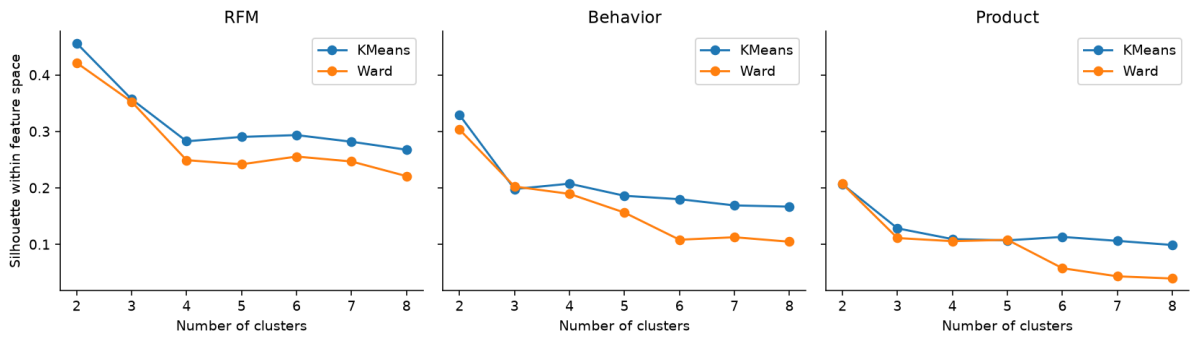


Figure 2: Candidate K-means and Ward silhouette curves. Compare cluster counts and algorithms within a panel; each panel defines its own distance space. Adding dimensions can alter apparent compactness even without a deterioration in practical usefulness.

silhouette 0.162 on retained rows. At tighter settings it yields six groups and negative silhouette. In Behavior, the only two-group setting has a group of just eight households and 14.2% noise. All tested Product settings produce one non-noise cluster, making multi-cluster silhouette undefined. These outcomes are reported rather than interpreted as zero scores. They apply to the tested radii and do not exclude better density-based specifications.

7.2 Stability and partition agreement

Table 4: ARI stability. Subsample columns summarize 20 paired independent 80% samples; reported ranges are empirical minima and maxima, not confidence intervals.

	K-means subsamples		Ward subsamples	
Space	Mean	Range	Mean	Range
RFM	0.980	0.948–1.000	0.535	0.011–0.874
Behavior	0.970	0.923–0.995	0.457	0.154–0.740
Product	0.964	0.860–0.990	0.314	−0.024–0.584

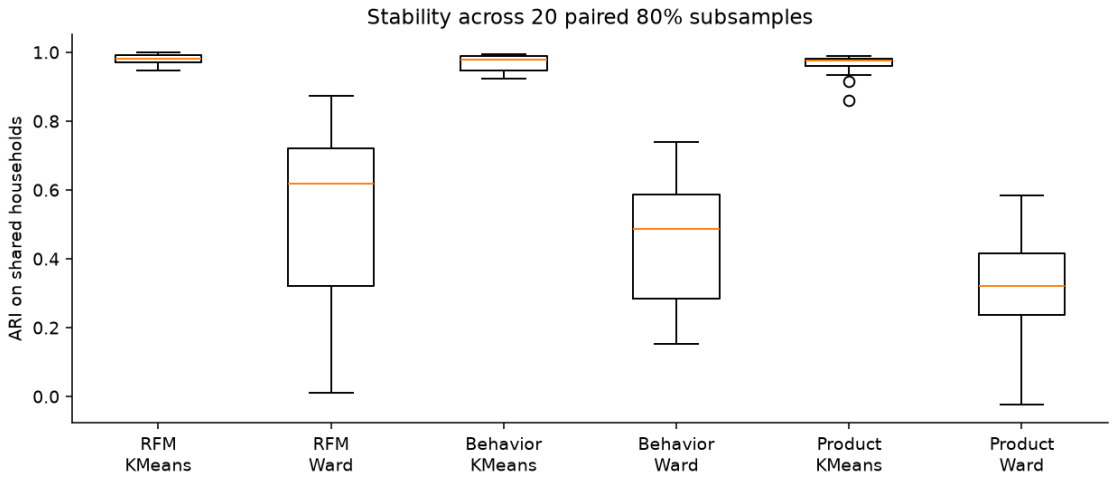


Figure 3: Conditional sampling stability on shared households. The K-means partitions are substantially more reproducible than Ward under this perturbation. Fixed preprocessing means these distributions do not measure uncertainty in the entire feature-construction pipeline.

Mean one-initialization ARI against the selected K-means fit is 0.994 for RFM, 0.982 for Behavior and 0.997 for Product. The principal findings are therefore not a consequence of one anomalous random start. K-means sampling stability remains high in all three spaces; Product's minimum subsample agreement is lower, so the additional representation does not improve robustness under this test. Ward is substantially more sensitive to household sampling, especially in Product space. Given the tested design, K-means is the more reproducible summary.

Across selected partitions, RFM–Behavior ARI is 0.614, RFM–Product is 0.589 and Behavior–Product is 0.850. Enrichment changes some assignments, and Product is much closer to Behavior than RFM. Assignment change is neither intrinsically good nor bad: it must be linked to new useful profile distinctions. Numeric cluster identifiers are space-specific; Product cluster 0 corresponds to the higher-activity group, whereas the higher-activity RFM and Behavior group is numbered 1.

8 Profiles and later-period descriptive checks

Table 5: Calibration profiles in original units. Means are descriptive household averages. H and L denote higher- and lower-activity groups within a space, not common fixed memberships.

Space	Group	n	Recency (days)	Baskets	Recorded sales
RFM	L	882	67.9	20.8	507.7
RFM	H	1,616	6.2	104.4	3,034.0
Behavior	L	809	68.9	23.8	418.1
Behavior	H	1,689	8.4	99.4	2,967.7
Product	L	843	65.8	25.5	448.4
Product	H	1,655	8.7	100.1	3,004.7

All representations chiefly separate higher-intensity, recently active households from less-frequent and less-recent households. In Product, higher-activity households average 549.7 distinct products and 12.8 departments, versus 114.1 products and 7.6 departments in the lower-activity group. They also have larger mean baskets (35.6 versus 23.0 sales units). Discount-line proportions are close (50.4% versus 48.9%), so these two groups should not be marketed as sharply distinct discount-response types.

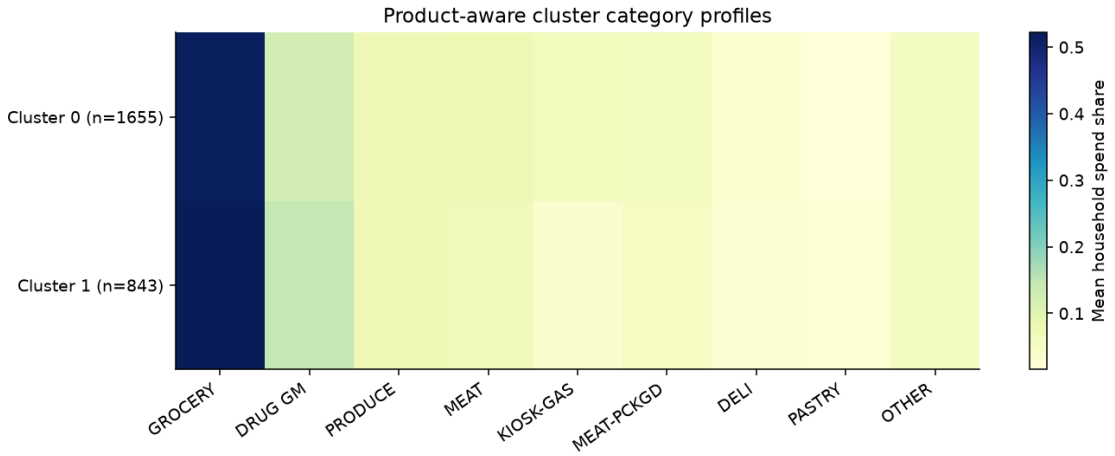


Figure 4: Product-aware department profiles: mean of household spend shares, rather than a share of pooled group revenue. Grocery dominates both groups; the modest contrasts qualify how much preference-specific structure the two-group solution reveals.

Both Product groups allocate about half of spending to Grocery (51.4% in the higher-activity group and 52.2% in the lower). The higher-activity group has a greater Kiosk-Gas share (6.0% versus 3.7%) and

Meat share (7.5% versus 6.6%), while the lower-activity group has more Drug GM (14.5% versus 12.3%) and Pastry (2.4% versus 1.6%). These are useful descriptions but modest absolute differences. They do not establish distinct dietary identities, price sensitivity, or promotion responsiveness.

Table 6: Later-period positive-purchase sales by frozen group. Brackets are 95% percentile bootstrap intervals for group means, conditional on the fixed clustering; activity is the proportion with any observed later purchase.

Space	Group	Mean later sales [interval]	Later active	η^2
RFM	L	343.2 [318.3, 371.2]	93.9%	0.204
RFM	H	1,487.5 [1,424.0, 1,547.0]	99.7%	
Behavior	L	306.3 [280.7, 332.1]	93.7%	0.197
Behavior	H	1,455.7 [1,393.7, 1,516.8]	99.5%	
Product	L	320.8 [297.6, 347.1]	94.1%	0.202
Product	H	1,471.9 [1,413.7, 1,535.6]	99.5%	

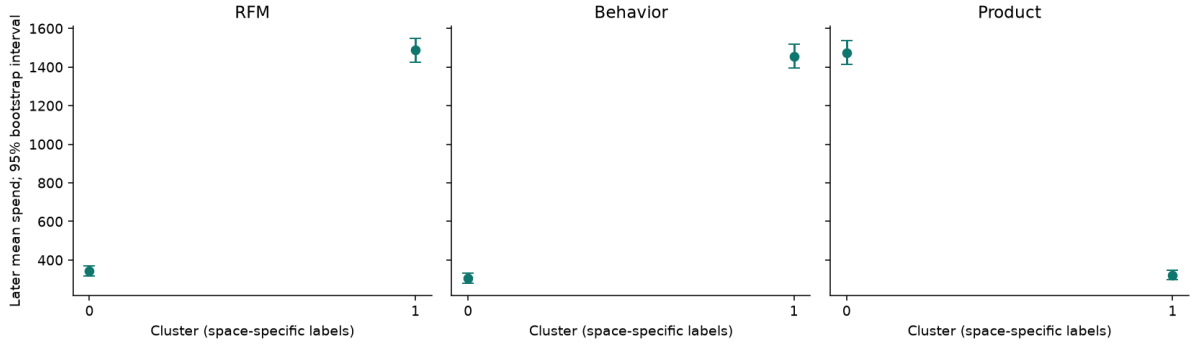


Figure 5: Later mean spending and bootstrap uncertainty. Calibration-defined groups remain descriptively different, with much larger within-cohort purchase intensity in the higher-activity group. Labels are independent in each panel.

The frozen groups retain a large later-period spending difference. Nevertheless, Product explains 20.2% of later spend variance, compared with 20.4% for RFM and 19.7% for Behavior. The absolute Product–RFM difference is approximately -0.002 in η^2. There is no descriptive improvement on this criterion. Most variance remains within groups; a segmentation is a coarse summary, not a precise individual-level spending forecast.

The intervals reflect group-mean uncertainty, not the uncertainty of a difference between representations. The models use the same households, so their metrics are statistically dependent. A paired household bootstrap of Product-minus-RFM explained variance gives -0.00190, with a conditional 95% interval $[-0.01426, 0.00860]$. It resamples the same household rows for both frozen assignments, preserving their dependence. This provides no clear evidence of improvement. No test of business lift or formal equivalence is claimed. Holdout results were not used in the within-space selection rule.

The notebook also includes a two-dimensional PCA projection. It is an illustrative view, not independent evidence of cluster separation.

9 Who changes groups, and why?

A change in membership is easiest to understand at household level. Cluster numbers themselves cannot be compared across independently fitted models. We first match Product labels to RFM labels by maximum household overlap, then inspect households whose matched assignments differ. From each direction of change, we select the household nearest its transition group's centroid in the transformed Product feature space, breaking distance ties by household identifier. The selection uses calibration data only; it does not seek dramatic future outcomes. The comparison concerns the full enrichment from RFM to Product, including basket behavior; it cannot attribute every change to department preferences alone.

After alignment, 287 of 2,498 households (11.5%) change groups: 163 move from RFM's lower-activity group to Product's higher-activity group, and 124 move in the reverse direction. These are cross-model reassignments of the same calibration households, not changes over time.

Household 2389: broader shopping despite less recent activity. This household last purchased 33 days before the reference day, made 43 baskets and recorded 1,102.61 in sales. Its 342 products across 13 departments accompany a move from the lower RFM group to the higher Product group. Meat accounts for 12.9% of spending and Produce 7.8%. The richer representation incorporates breadth and basket behavior that RFM does not explicitly encode.

Household 1970: recent visits with smaller baskets. With recency of seven days and 44 baskets, this household moves from the higher RFM group to the lower Product group. Recorded sales total 615.01; mean basket sales are 13.98, compared with 25.64 for household 2389. Grocery and Drug GM account for 66.5% and 17.9% of spending, with 214 products across eight departments. Similar visit counts therefore need not imply similar basket scale or mix.

These are representative examples under the stated distance rule, not causal explanations of individual assignments. Both the basket variables and department block change between RFM and Product; the comparison does not isolate the effect of one variable.

10 Do department differences persist later?

The original later-spending criterion is naturally related to RFM's monetary feature. We therefore add a category-specific descriptive check: do the frozen groups distinguish later department spending shares? This extension was designed after the original holdout results had been inspected. It is explicitly exploratory, and it does not change the original feature fitting, cluster-count selection or frozen memberships.

The category vocabulary remains the eight largest calibration departments plus Other. Later purchases in any department outside that vocabulary enter Other. We calculate each household's department shares separately and give each household equal weight. Shares are undefined for the 59 calibration households with no positive later purchase: they remain in the full-cohort spending and activity results, but the category-composition comparison uses the same 2,439 active households for every representation. This separates absence of purchases from preference for a department.

For household share vector $\mathbf{s}_i$ and frozen group g, the category separation measure is

$$\eta^2_{\text{category}} = \frac{\sum_g n_g \|\bar{\mathbf{s}}_g - \bar{\mathbf{s}}\|^2}{\sum_i \|\mathbf{s}_i - \bar{\mathbf{s}}\|^2}.$$

It is the fraction of total squared variation in nine-component household shares that lies between group means. This raw-share Euclidean summary weights departments by their observed variation and is not a universal measure of segmentation quality. We compare Product minus RFM using 1,000 paired household bootstrap replicates (seed 2027), keeping fitted memberships fixed. The percentile interval excludes pipeline-refitting uncertainty and is exploratory rather than a confirmatory test.

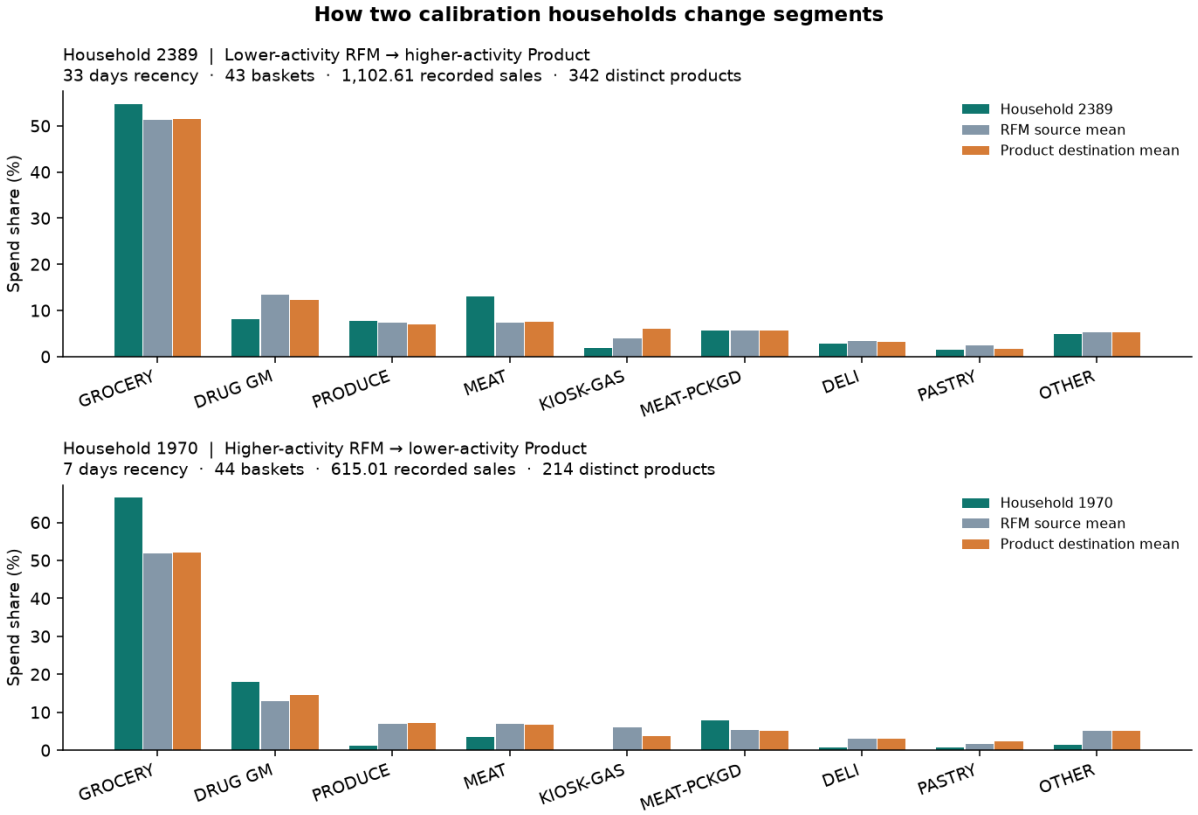


Figure 6: Actual calibration households selected by a reproducible rule from the changed-membership groups. Sales and basket counts describe the calibration period only. These cases illustrate how different definitions of similarity change a household's assignment; they are not inferred customer identities or evidence that either assignment is correct.

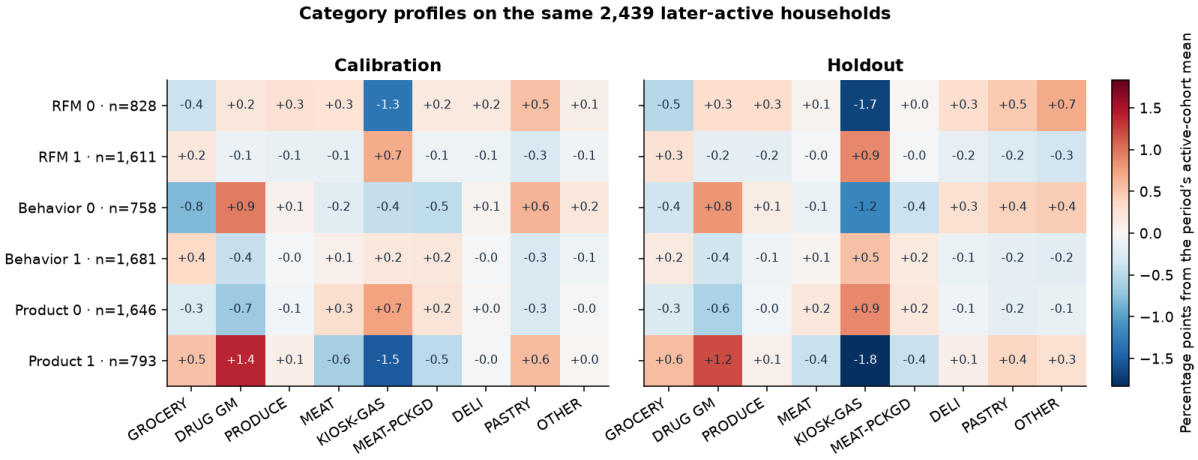


Figure 7: Department-share deviations for frozen groups. Calibration and later profiles use the same later-active household cohort, with each period centered on that cohort's mean household share. A repeated direction of deviation suggests descriptive persistence; it does not establish a response to marketing.

Table 7: Exploratory category-share separation on the identical 2,439 later-active households. Entries are percentages of total squared share variation explained by frozen group means, not percentages of shoppers or sales.

Representation	Calibration	Later period
RFM	0.250%	0.299%
Behavior	0.221%	0.180%
Product	0.529%	0.386%

Product has the largest point estimate for later category separation, but it explains only 0.386% of the total variation in household shares, versus 0.299% for RFM. The Product-minus-RFM difference is 0.000868 on the fraction scale (0.087 percentage points), with a conditional 95% interval of $[-0.001097, 0.003148]$. The interval spans zero, so the extension does not establish a clear improvement. Large within-group preference variation remains.

Some profile directions do persist. Among the later-active cohort, Product's lower-activity group is 1.39 percentage points above the calibration cohort mean for Drug GM and 1.20 points above the later mean. Its Kiosk-Gas deviations are -1.53 and -1.83 points respectively. Persistence is not universal: its small Deli deviation changes sign. These patterns support a limited descriptive interpretation, not sharply separated shopper personas or a marketing policy.

What this adds to the decision. Even when evaluated against department preferences rather than later spending, the selected two-group Product model shows only a small, uncertain advantage. Retain RFM for broad activity monitoring; use category profiles as supplementary descriptions unless a separately validated category-specific decision justifies a more complex segmentation.

11 Fixed-count ablation and sensitivity

The selected two-group fits already hold cluster count constant. A second paired comparison fixes $k = 3$ to ask whether enrichment reveals more useful structure at a moderately finer resolution. This is a supplementary analysis, not a new selection rule tuned to later results. At three clusters, RFM retains

Table 8: Paired K-means comparison at $k = 3$. Common-space silhouette evaluates every assignment in the same standardized RFM coordinates; it tests preservation of RFM geometry, not a neutral measure of all possible usefulness.

Space	Smallest group	Common RFM silhouette	Mean subsample ARI	Later η^2
RFM	397	0.357	0.970	0.313
Behavior	614	0.138	0.763	0.215
Product	677	0.079	0.929	0.170

stronger sampling reproducibility than either extension and explains a larger fraction of later-spend variance. Product is more stable than Behavior in this finer partition, but still offers no later-spend separation improvement over RFM. The paired bootstrap Product-minus-RFM difference in later η^2 is -0.143, with a conditional 95% interval $[-0.165, -0.119]$. This is weaker stratification for the spending criterion; it does not evaluate category-specific marketing value. The common-space scores show that the enriched assignments depart from the baseline's geometric criterion, as expected when the definition of similarity changes. They cannot show that category distinctions are intrinsically wrong.

The Product representation's block weight is consequential. At fixed $k = 3$, multiplying its category coordinates by 0.5 produces ARI 0.433 against the equal-block model; multiplying them by 2 produces ARI 0.460. Because variance scales with the square of a coordinate multiplier, these settings change

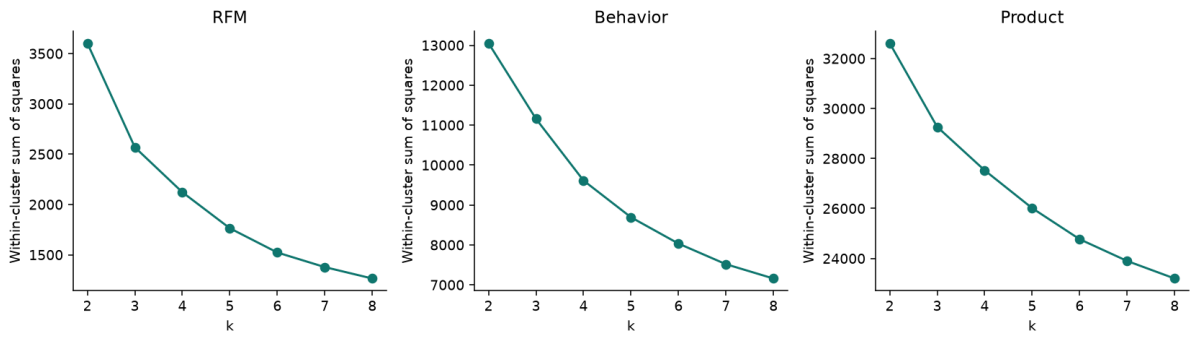


Figure 8: K-means objective over the candidate cluster counts. Read each elbow within its own representation. Absolute inertia values across spaces have different scales and are not a ranking of model quality.

the category block's variance to one quarter and four times its default. Such changes lead to materially different memberships even though all use the same data. This sensitivity is a stronger qualification than the high repeated-start stability: an optimizer can find a reproducible answer to a representation choice that remains debatable.

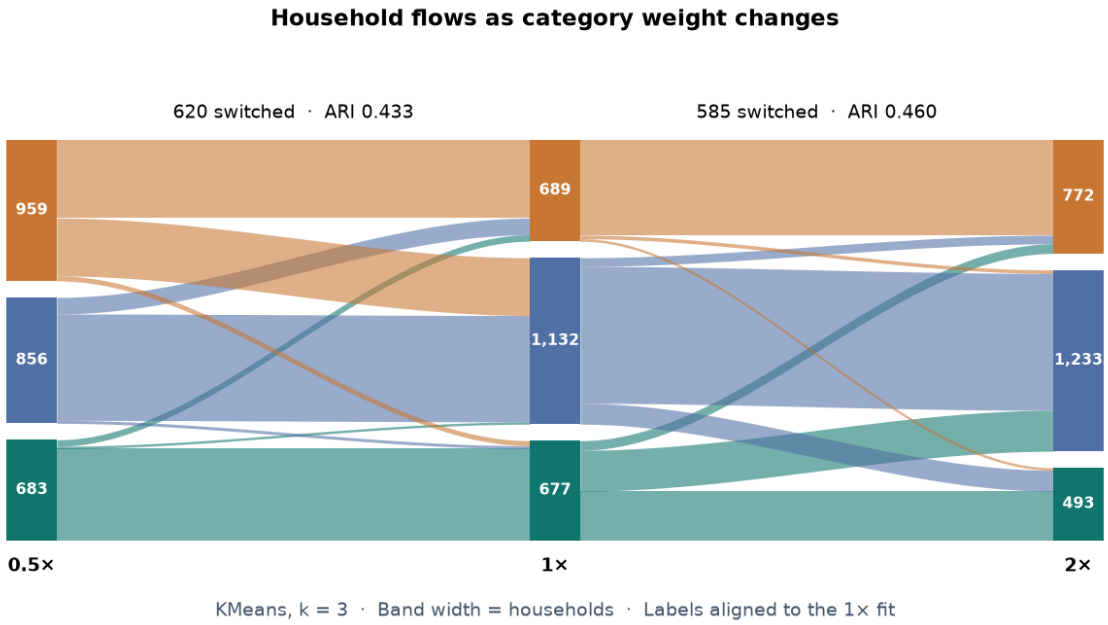


Figure 9: Household membership at three category-coordinate weights, holding $k = 3$ fixed. Ribbon widths count households. Nondefault labels are matched to the default partition by maximum overlap; color is a visual correspondence, not a claim of identical cluster meaning. ARI remains invariant to label numbering.

After maximum-overlap label alignment, halving the default category-coordinate weight changes 620 of 2,498 memberships (24.8%), while doubling it changes 585 (23.4%). These counts compare each alternative with the default; they are not temporal migration rates.

The flow view separates two questions: whether the optimizer repeatedly finds the same answer at a fixed weight, and whether that answer survives a different definition of similarity. Strong initialization stability answers the first question only. A practical segmentation should disclose the weighting choice and inspect the households affected by it.

12 Practical interpretation and limitations

12.1 What the evidence supports

For this cohort, time boundary, feature construction and evaluation suite, the RFM baseline is the preferred default summary. It is parsimonious, reproducible under household resampling, and at least as informative for later spending separation as the tested extensions. The evidence supports two broad activity tiers, not an assertion that exactly two immutable consumer types exist. Product-aware features add interpretable department descriptions and change household membership, but do not demonstrate an improvement in the principal diagnostics. The exploratory later-category check also finds only a small, uncertain advantage: 0.386% versus 0.299% of share variation explained, with a paired difference interval spanning zero.

A retailer could use these groups to organize descriptive monitoring: compare assortment coverage, observed return activity and basket patterns between higher- and lower-activity households, and investigate whether a subgroup has unmet needs. These are hypotheses for subsequent research. A recommendation to send a particular coupon, suppress an offer, or prioritize a household would require an explicit decision objective, operational constraints and an experiment measuring incremental outcomes. The present study does not estimate coupon responsiveness, campaign return, margin or customer lifetime value.

12.2 Limits of the data and design

The data concern a selective historical retailer panel and do not capture total household grocery demand. Household behavior can be influenced by panel enrollment, incomplete observation, seasonality and store assortment. Positive-purchase filtering omits returns and zero-sales lines, so monetary features represent recorded buying activity rather than net margin. Historical transaction records do not automatically generalize to current shoppers or another market.

Only one temporal split is examined. A 500-day calibration aggregates long-run intensity and may blur changing preferences; the 211-day holdout can have a different seasonal mix. No observed later purchase does not prove defection. Future spend is only one descriptive relevance criterion and naturally favors historical spending features. Product-aware groups might matter for a different prespecified objective, such as assortment breadth or category demand. The exploratory category-share extension addresses later descriptive preference separation, but still does not establish operational value.

The candidate grid is limited, DBSCAN tuning is exploratory, and only standard Euclidean K-means and Ward variants are tested. Correlated behavior features and category block weighting affect distances. Composition is coarse, restricted to eight departments plus Other, and fine category patterns can be hidden by aggregation. The $k = 3$ sensitivity check shows that weighting uncertainty materially changes membership. A PCA picture loses dimensions and should not be used as proof of separability.

Stability checks condition on fixed preprocessing and do not refit the choice of top departments or transformations. Twenty resampling repetitions provide a useful diagnostic, not a population confidence interval or proof of a true partition. Bootstrap later-spend intervals condition on fixed fitted groups. There is no independent ground truth, external replication, formal model-equivalence test or causal experiment.

12.3 Next steps

A stronger follow-up would predefine the retail decision and a relevance metric independent of the feature-development process, repeat the analysis over rolling temporal origins, and refit the entire pipeline within resampling. It would compare multiple category resolutions and prespecified composition weights, extend the household case studies to systematic transition analysis, and monitor membership drift. A category-specific objective could determine whether preference information adds value beyond intensity. Randomized trials would be needed to evaluate targeting benefits and costs.

13 Conclusion

This audited, reproducible comparison finds no demonstrated improvement from the tested enrichment beyond RFM. All representative K-means fits select two broad activity groups and are more reproducible than Ward. RFM remains the preferred default: its mean subsample ARI is 0.980 and later-spend explained variance is 0.204, versus Product's 0.964 and 0.202. The fixed three-group comparison also favors RFM for later spending separation. Product features add department descriptions, but membership changes materially with category weighting. The exploratory later-category analysis finds some persistent profile differences, yet no clear improvement over RFM in overall category-share separation. These observational findings support a restrained baseline recommendation; a different retail objective or a prospective experiment is needed to establish incremental business value.

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